Siddharth Anand

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Education

2025 – Present **Purdue University, West Lafayette, USA**
M.S. *Aeronautics and Astronautics*
Major *Autonomy and Control*

2020 - 2025 **Indian Institute of Technology Bombay, India** 8.44 CPI
Masters *Control and Computing*
Bachelors *Electrical Engineering*
Minor *Computer Science*

2018 - 2020 **GVN - The Global School, India** 94.6%
Intermediate/+2

2008 - 2018 **Delhi Public School, India** 94.8%
Matriculation

Interests

Perception-Driven Autonomy | State Estimation | UAV Systems | Simulation-Based Robotics

Key Projects

Vision-Based Multi-UAV Tracking and State Estimation *Jan '26 - Present*
Prof. James M. Goppert Graduate Research Assistant
Purdue UAS Research and Test Facility (PURT)

- Developing a **Multi-Camera, Multi-Object Vision-Based Perception Pipeline** for UAVs, processing distributed camera detections for real-time multi-agent tracking
- Implementing **Probabilistic State Estimation** using **Particle Filters** to fuse uncertain multi-camera detections, enabling reliable 3D trajectory estimation from noisy visual observations
- Integrating **ROS 2** and **Gazebo-based simulation** for system-level implementation

Anti-Drone and Counter Anti-Drone Systems *July '24 - July '25*
Prof. Debraj Chakraborty Dual Degree Project

- Developing a Pursuer-Evader Algorithm for autonomous drones, focusing on Anti-Drone Strategies, using **Airsim** as the control framework and **Unreal Engine** for realistic environment simulation
- Enabled **real-time object detection** and tracking via drone-mounted cameras within Unreal Engine and integrated **Adaptive Camera Guided Control (ACGC)** to maintain the evader within the field of view during pursuit
- Incorporating a **3D Voronoi-based Pursuit Strategy** to optimize drone trajectories by dynamically partitioning the environment, enabling real-time coordination of multiple pursuer drones

Active Learning and Core-Set Selection *January - April '24*
Prof. Amit Sethi Supervised Research Exposition

- Implemented the Core-Set approach to active learning using **VGGFeat** as the backbone architecture, optimizing CNN performance on CIFAR-10 and **CRC-VAL-HE-7K** image datasets
- Achieved significant improvements in annotation efficiency and model accuracy, with CRC-VALHE-7K dataset results showing test accuracy of 81.314% for **batch size 256** after 50 epochs

AI Engineer Intern *May - July '24*
HireQuotient

- Developed and deployed an **AI-Driven Chatbot** using **OpenAI Agents** to handle user queries, integrating and uploading company documentation to a **Vector Database** for reliable responses
- Built an **Automated Outreach System** with **Gen AI**, allowing users to Delete, Enrich Contact Details, and Trigger Outreach to candidates via prompts, while tracking these actions in **MongoDB**
- Implemented **customized Email notifications** to communicate the results in the outreach system
- Led **Backend Development** with **JavaScript**, managed **APIs on GitHub**, and integrated multiple APIs to support the AI agent's functionality, including exploring **AI Agent function calling**

Unmanned Aerial Vehicle (UAV) Technology *May - July '22*
ideaForge Technology | RnD: Software and Control Intern Guide: Prof. Debraj Chakraborty

- Developed and implemented a gimbal system using **Model-Based** techniques and **Reinforcement Learning** algorithms to stabilize Pan-Tilt-Zoom camera payloads on UAVs for improved stability
- Applied **Line-of-Sight Optimization** techniques to improve the performance of drone mapping

Deep Learning and Image Segmentation *July - November '22*
Prof. Amit Sethi — Image Processing Course Project

- Successfully solved the **MoNuSeg Challenge** using Deep Learning techniques, specifically by constructing a **UNet Architecture** in TensorFlow and Keras for semantic segmentation
- Trained the UNet Model using a combination of **Dice Loss** and **MS-SSIM** with a trade-off parameter, also implemented the **Watershed Segmentation** algorithm for comparison
- Applied Machine Learning techniques, specifically a **Support Vector Machine** with **RBF Kernel**, for image enhancement and achieved optimal performance through hyperparameter tuning

Data Engineering and NLP Application *May - July '23*
rupyy by CarDekho Group | Cloud Computing Intern

- Developed **Extract Transform Load (ETL) Pipeline** in Python to transfer loan data from RDS to S3, implementing both batch processing and streaming solutions using **AWS Lambda**
- Utilized **LLM FlanT5** to generate answers based on PDFs to build a query response systems

Team Veloce Hyperloop IITB *October '21 - October '22*
Student technical team participating in SpaceX's Hyperloop Pod Competition

- Assisted in the design and development of the **Control Architecture** for the Hyperloop Pod, including the integration of sensor fusion and communication protocols
- Selected as one of the top teams to present in the Stage-1 documentation review of the **European Hyperloop Week**, showcasing the progress and potential of the Hyperloop Pod control system

Depth Prediction in Single Monocular Images *March - April '22*
Prof. Abir De | Introduction to Machine Learning Course Project

- Developed a **Multi-Scale Deep Neural Network** to accurately predict the depth of single monocular images with improved accuracy and efficiency
- Utilized a **Coarse-Scale Network** to handle the task at a global level and a **Fine-Scale Network** to refine the predictions at a local level, resulting in high-precision depth predictions
- Evaluated the model on the NYU-Depth Dataset, achieving results in terms of **Scale-Invariant Loss** (0.221 in log scale) and **L2 Loss** (0.285 in log scale)

Stock Market Analysis *May - July '21*
Summer of Science | Maths and Physics Club

- Learned the basics of **Equity**, **Fundamental Analysis**, **Technical Analysis** of company stocks and started trading to apply strategies learned and gained experience about entry and exit points
- Implemented practical applications by shortlisting mid-cap stocks in **Real-World Stock Market**

Scholastic Achievements

- Awarded **Valedictorian**, Control and Computing Group, EE Dept., IIT Bombay *('25)*
- Selected as one of two Indian students for *semester exchange* at **KAIST University** *('23)*
- Achieved **99.77 percentile** in *JEE Advanced* amongst 150,000 candidates *('20)*
- Achieved **99.28 percentile** in *JEE Main* amongst 0.9 Million candidates *('20)*
- Qualified **Regional Mathematics Olympiad** (RMO) conducted by HBCSE *('19)*
- Recipient of prestigious **National Talent Search** (NTS) Scholarship *('18)*

Position of Responsibilities

Newsletter Editor *May '21 - May '22*
Official Student Newsletter | Dept. of Electrical Engineering

- Served as a **Lead Panelist** for Background Hum, a technical publication, overseeing the development of articles and ensuring their technical accuracy and relevance
- Mentored junior panel members and helped them develop their writing and research skills

Technical Skills

Programming	Python | C++ | JavaScript | MATLAB
Simulation & Robotics	Unreal Engine | AirSim | ROS | GNU Radio
Machine Learning	TensorFlow | PyTorch | Keras | Pandas | Matplotlib
Tools & Others	Git | LaTeX | AutoCAD | VHDL

Courses Undertaken

Control & Computing	Lie Group Methods*, Estimation and Identification, Nonlinear Dynamical Systems, Multivariable Control Systems, Optimal Control Systems, Linear Systems, Control Systems (with Lab), Estimation Lab
Machine Learning	Introduction to ML, Image Processing, Distributed Optimization & ML
Mathematics	Matrix Computations, Game Theory, Applied Linear Algebra, Probability and Random Processes, Aerospace Engineering Probability & Estimation
Aerospace & Related	Space Flight Dynamics, Advanced Methods in Satellite Image Processing
Others	Technology Entrepreneurship And Research Translation

*Currently enrolled courses at Purdue University

Miscellaneous

- Built an **Obstacle Sensing** and **Maneuvering** remote-controlled bot in XLR8 *('21)*
- Successfully completed **National Sports Organization** course of **Athletics** at IIT Bombay *('20)*
- One of the youngest of nearly **300** delegates at the **International Model United Nations** *('17)*
- Achieved **Top Honors** in the **UCMAS** Abacus program, graduating from all 8 levels *('14)*